

Learning a shortcut to gravitational-wave inference

A neural approximation to an expensive likelihood could make gravitational wave analysis faster and more adaptable. Its usefulness depends on preserving the uncertainties that turn a detected signal into a physical measurement.

Gravitational waves (GW) are ripples in spacetime intrigued by movement of massive objects such as black holes or neutron stars. Decade after the first observation of GW plenty of signals are thereafter recorded, meanwhile more complicated theoretical models have been put forward to investigate more phenomena even the beyond general relativity ones. All these work relies on proper parameter estimation (PE) which is highly demanding in computational resources. Negri and Samajdar¹ develop an algorithm using neural likelihood to significantly accelerate PE and for the first time apply such algorithm on a real GW signal.

Gravitational waves carry information about their sources, but detectors record a noisy strain data rather than the parameters themselves including mass, angular momentum and direction etc. Recovering these properties requires comparing the data with theoretical waveforms determined by them, acquiring probability distributions of parameters.

Such recovery process is Bayesian Inference, the information of parameters of interest is encoded in the posterior probability density function:

$$p(\vec{\theta} | H_s, d) = \frac{\mathcal{L}(d | \vec{\theta}, H_s) p(\vec{\theta} | H_s)}{\mathcal{Z}}. \quad (1)$$

Here d is for the exact data recorded by detectors, $\vec{\theta}$ for parameters of interest, and H_s stands for the hypothesis that data contains GW signal. The likelihood $\mathcal{L}$ measures how well the data is explained by proposed parameters. In the GW detection case, it is calculated indirectly under the assumption that noise adhere to Gaussian distribution. This process includes cumbersome integrals and is challenging in computation. Additionally the high dimensionality of parameters requires $\mathcal{O}(10^6 \sim 10^8)$ likelihood points to depict the whole distribution. These factors together make the conventional Bayesian PE computationally expensive.²

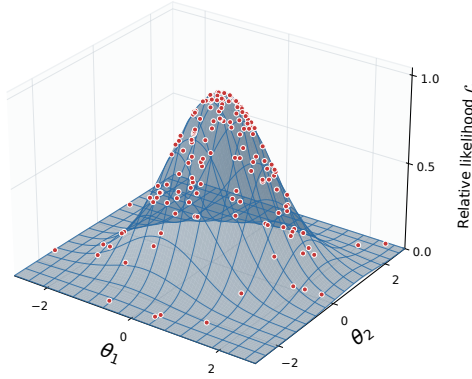
Significant progress has been made in faster and cheaper PE, existing approaches features by skipping the burden by a likelihood-free inference, training neural network to estimate posterior directly, reducing the computational costs in evaluating likelihood directly.^{3;4;6} Their differing training requirements and validation strategies leave room for complementary approaches.

The core concept of Negri and Samajdar¹'s PE algorithm is based on neural network (NN)'s ability of approximating any mapping. An NN is trained as a mapping $[\vec{\theta}_i] \rightarrow [\mathcal{L}(\vec{\theta}_i)]$ that resembles the true likelihood function based on selected training points of true likelihood evaluation (Fig. 1a). Once trained, NN estimates likelihood values through a short sequence of weighted sums and simple nonlinear operations, bypassing the expensive true likelihood evaluation.

Afterwards, NN could fit in Bayesian PE as replacement of true likelihood function. The posterior is then acquired by stochastic sampling across parameter space for NN likelihood estimation and reviewed by a metric to assess its accuracy in approximating true posterior. If examination failed, a new training cycle is restarted, with sampled posterior points and their true likelihood being added into the training set. This allows the network to correct itself from inaccurate regions. Fig. 1b illustrates the improvement: a spurious peak at low mass ratio in the first cycle disappears with further training, while the recovered distributions approach those obtained using the original likelihood.

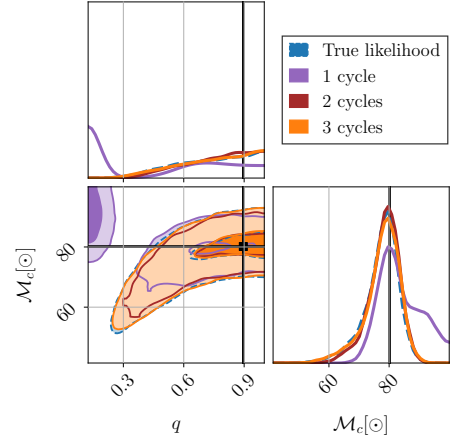
a Learning the likelihood

— True likelihood
 — Neural approximation $NN(\theta)$
 • Training points: exact evaluations



Conceptual toy example; relative likelihood, not a normalized parameter density.

b Improving the posterior



Source: Negri & Samajdar (2025), Fig. 4(b).

Figure 1: Learning and refining a likelihood approximation. **a**, In a conceptual two-parameter example, training set consists of true likelihood evaluations (red dots) and constrain a neural network approximation (blue net surface) to a true likelihood surface (grey). Points of training set concentrate near the likelihood peak while also covering the outskirts; the height is relative likelihood. **b**, Successive training cycles improve posterior agreement with a reference analysis for parameters mass ratio and chirp mass. Subfigure b is reproduced from Fig. 4(b) of Negri and Samajdar.¹

Simulated signals test whether the method preserves meaningful uncertainty estimates. The reported checks broadly support its calibration within the tested regime, while also identifying features that warrant closer investigation.

For GW150914, FLEX reproduces the main parameter constraints at substantially lower computational cost. Tests with several waveform models, together with reuse of the trained likelihood in another sampler, illustrate its practical flexibility.

The likelihood surrogate also permits evidence calculations for comparing hypotheses, but agreement between posteriors does not by itself establish evidence accuracy. Multimodal distributions and more demanding signals remain important tests of the approach.

Cheaper inference could make repeated physical tests more accessible as gravitational-wave observations become richer. The next challenge is to retain this efficiency while demonstrating reliable uncertainties across a wider range of sources and analysis choices.

References

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